



ASSIGNMENT - 04

Supervised Machine Learning

DATA SCIENCE

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SUPERVISED MACHINE LEARNING

1. What is supervised learning?
2. How does supervised learning differ from unsupervised learning?
3. Explain the difference between classification and regression.
4. What is overfitting in a machine learning model?
5. How can you prevent or reduce overfitting? Name two techniques.
6. What is the bias-variance tradeoff in machine learning?
7. What is cross-validation and why is it used?
8. What is a confusion matrix? What do the entries (TP, FP, TN, FN) represent?
9. Define precision, recall, and F1-score. Why might precision-recall be more important than accuracy in some cases?
10. What metrics would you use to evaluate a regression model's performance?
11. What is linear regression and what assumption does it make about the relationship between the variables?
12. How is logistic regression different from linear regression? What type of problems does logistic regression solve?
13. What is a decision tree and how does it make decisions?
14. In a decision tree, what are entropy and information gain? How are they used?
15. What is a random forest and how does it improve upon a single decision tree?
16. What is the difference between bagging and boosting methods in ensemble learning?
17. What is gradient boosting? Name an example of a gradient boosting algorithm.
18. What is a Support Vector Machine (SVM)? What kind of decision boundary does it try to find?
19. What is the "kernel trick" in SVMs and why is it useful?
20. What is a Naive Bayes classifier and why do we call it "naive"?
21. What is regularization in machine learning, and what is the difference between L1 and L2 regularization?
22. What is the difference between a model parameter and a hyperparameter? Give examples of each.
23. What is gradient descent and how is it used in training models (e.g., in logistic regression or neural networks)?
24. What are training, validation, and test sets, and how are they used in model development?
25. How do you handle imbalanced classes in a classification problem?
26. What is the k-Nearest Neighbors (kNN) algorithm and how does it make predictions?
27. What is an ROC curve and what does the area under the curve (AUC) represent?
28. What is the "curse of dimensionality"?
29. When faced with a new dataset, how do you decide which machine learning algorithm(s) to try? What factors influence your choice?
30. What is semi-supervised learning?